BRANDON PAUTLER

Kingston, ON | bpautler@bell.net | +1 548 888 8801 | www.linkedin.com/in/brandon-pautler

EDUCATION

Queen's University **Kingston, ON**
Bachelor of Applied Science (BASc) in Electrical Engineering *Sep 2022 – Apr 2027*
- Relevant Coursework: Power Electronics, Integrated Circuits, Electronics, Electromagnetics, Microprocessor Systems

EXPERIENCE

Ontario Power Generation **Pickering, ON**
System Engineering – Reactor Control Intern *May 2025 – Jun 2026*
- Diagnosed faulty **RTD temperature readings** in the **Reactor Regulating System (RRS)**, a safety-critical control system, by analyzing trend data and tracing system drawings; recommended a corrective replacement that restored normal readings
- Built custom **data analysis tools in Seeq** to surface plant conditions not directly visible in available data, supporting investigations across RRS and one of the emergency shutdown systems
- **Restored Cobalt-60 processing equipment** after a failure forced manual operation and then halted processing; sourced a qualified substitute for an obsolete part and planned the replacement with maintenance
- Analyzed **failure history** of submerged, radioactive equipment, leading to a new preventive maintenance work order
- Performed **20+ field walkdowns**, 10+ inside the reactor building under radiological controls, inspecting malfunctioning equipment and taking field readings for troubleshooting

Robarts Research Institute, Western University **London, ON**
Student Intern *Summers 2023 & 2024*
- Owned **hardware and control design** on a 3-person team building a 3D-printed **Fourier ptychography microscope** (~$150 CAD in parts) for malaria diagnosis; led imaging tests that **resolved malaria parasites** in red blood cells
- Designed a **custom 2-layer LED illumination PCB** (~150 addressable LEDs, STM32 microcontroller, 3.3 V to 5 V level shifting) that mounts on a Raspberry Pi, refining it over **3 board revisions**
- Eliminated the flicker of off-the-shelf LED matrices that disrupted imaging, raised brightness, and added a ring LED layout shown in prior research to improve reconstructions
- Wrote **C firmware** (STM32) and a **Python library** (Raspberry Pi) for two-way SPI control of LED patterns and brightness, sequencing illumination in sync with camera capture
- Redesigned the 3D-printed microscope body in Onshape over 5+ iterations to fix lens, sensor and aperture misalignment; **calculated the aperture size** to admit light from the outermost LEDs and block stray reflections

PROJECTS

MRI-Compatible Touch Tablet
Capstone Project, team of 4 *Sep 2026 – Present*
- Developing a battery-powered touchscreen tablet that **operates safely inside an MRI scanner** for a Queen's psychology lab's fMRI studies, with video and touch data carried over fibre optics
- Adapting a commercial display for MRI use by **replacing ferrous components** on its driver board with non-magnetic parts
- Designing **custom air-core power inductors**, sized with Wheeler's formula, to replace ferrite-core parts that saturate in the scanner's field, where no commercial air-core part exists at the required µH values
- Designing a **3D-printed plastic enclosure** in CAD for the display, electronics and battery, with no metal fasteners or framing

Headphone Cable Repair
Personal Project *Feb 2025*
- Repaired a severed headphone cable by **soldering its three fine-gauge conductors** and **potting the joint in resin** in a custom 3D-printed mold that kept the connections isolated, then sanded it into a durable finished housing

PUBLICATIONS

- H. J. Yang, W. Wasswa, R. Li, **B. Pautler**, et al., "Optical resolution and MTF of a low-cost Fourier ptychography microscope using a Raspberry Pi computer," *Proc. SPIE* 12832, 1283205 (2024). doi:10.1117/12.3003269

SKILLS & INTERESTS

- **Hardware**: PCB Design (EasyEDA), SMD & Through-Hole Soldering, STM32, Raspberry Pi, SPI, Onshape, 3D Printing
- **Software**: Embedded C, Python, MATLAB, Assembly, LTspice, Seeq, Linux, Microsoft Office
- **Test Equipment**: Oscilloscope, Logic Analyzer, Spectrum/Network Analyzer, Function Generator, Power Supply, DMM
- **Interests**: Hiking, Oil Painting (Landscapes), Guitar, Leatherworking, Woodworking, Cooking, Ice Hockey, Golfing